MIL-F-19426D
29 May 1986
SUPERSEDING
MIL-F-19426C
19 August 1983

MILITARY SPECIFICATION

FORMING MACHINES, SHEET AND PLATE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers power driven sheet and plate forming machines with equipment and accessories specified herein.

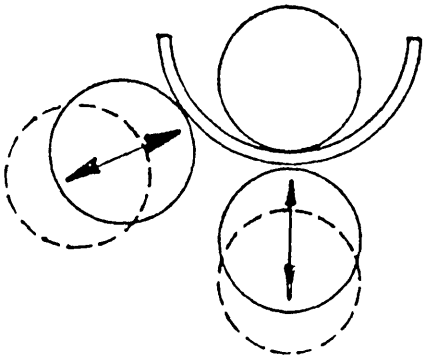
1.2 Classification. Forming machines shall be of the following types and the sizes listed in tables I, II, III, and IV. The type and size to be furnished shall be as specified (see 6.2.1). Figure 1 depicts the configuration of rolls for each type machine.

Type I - Initial-pinch, three roll (see figure 1).
Type II - Pyramid, three roll (see figure 1).
Type III - Pinch-pyramid, three roll (see figure 1).
Type IV - Double-pinch, four roll (see figure 1).

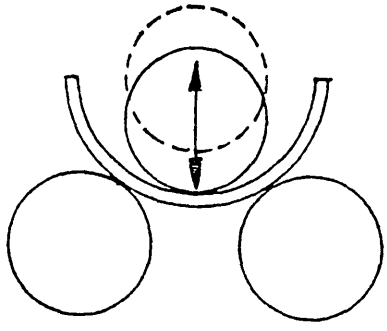
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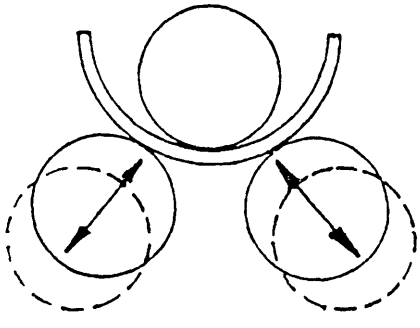
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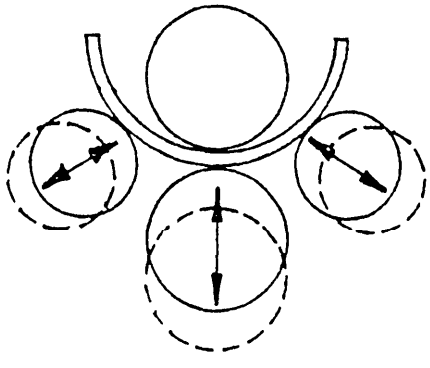
TYPE I
Initial-pinch, three roll



TYPE II
Pyramid, three roll



TYPE III
Pinch pyramid, three roll



TYPE IV
Double-pinch, four roll

FIGURE 1. Illustrations of roll position and adjustment.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitations

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B11.12 - Roll Forming and Roll Bending Machines, Safety Requirements for the Construction, Care, and Use.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the Publication Sales Division, National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

(Non-government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1) a sample shall be subjected to first article inspection (see 4.4 and 6.3).

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3.2 Design. The forming machine shall be new and one of the manufacturer s current models capable of bending steel sheets and plates to produce uniform cylinders and cylindrical segments in accordance with the requirements herein. Types I, II, and III machines shall have three rolls mounted in the horizontal plane and the type IV machine shall have four rolls mounted in the horizontal plane. The machine shall include all components, parts, and features necessary to meet the requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2.1 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, roll position indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.1.1 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and SI System of Measurements.

3.2.2 Reclaimed materials. The forming machine may contain reclaimed materials provided such materials will not jeopardize the intended use and performance of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.3 Energy efficiency. The machine and its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.4 Controls. All operating controls shall be located conveniently to the operator at his normal work station.

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3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Environmental protection. The machine shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at point of installation (see 6.4).

3.2.8 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible. When specified (see 6.2.1), a centralized automatic lubrication system shall be provided.

3.2.9 Interchangeability. All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The forming machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.32 Welding, brazing, or solering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

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3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, lash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachment and accessories shall conform to FED-STD-H28.

3.3.7 Dials. Dial diameters shall be such that graduations may be read from normal operating position. Dials shall be permanently and legibly engraved or etched on a nonglare background.

3.3.8 Gears. All gears shall be constructed from a material suitable for the intended purpose. All drive gears shall have the hardness, surface finish, and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear.

3.3.9 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The forming machine shall consist of not less than a base, a housing, a drive system, rolls, a tail hinge, an electrical system, and when applicable, a hydraulic system.

3.4.1 Base. The base shall be either cast iron, cast steel, or a fabricated steel weldment of a mass and weight that will support without permanent distortion, the end frames, rolls, drive system, and other components of the machine while operating within its rated capacity. The base shall have means to permit fastening the machine to the floor or foundation.

3.4.2 Housing. The housing shall be constructed to provide rigid support for all components when the machine is operated within its rated capacity.

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3.4.3 Drive system. The main drive system shall be either hydraulically driven, mechanically driven, or hydromechanically driven. Unless otherwise specified, the drive shall be variable speed. When specified (see 6.2.1), or when tables I, II, III, or IV indicate that a variable speed drive is not available, the main drive shall be single speed.

3.4.4 Rolls. All rolls shall be of solid construction and shall be made of steel with a high carbon content. The lower rolls shall have one or more longitudinal grooves for the full length of the roll. When specified (see 6,2.1), the lower rolls shall have one or more circular grooves for bending workplaces with wired edges.

3.4.4.1 Initial-pinch bending rolls. The initial-pinch, type I, forming machine shall have three rolls arranged in an asymmetrical triangle configuration. The top pinch roll position shall be nonadjustable. The lower front pinch roll shall be adjustable vertically to accommodate different plate thicknesses. The rear bending roll shall be adjustable angularly in relation to the pinch rolls to control the diameter or curvature of the workpiece. All three rolls shall be power driven.

3.4.4.2 Pyramid three roll. The pyramid, type II, forming machine shall have three rolls arranged in a symmetrical pyramid configuration. The upper roll shall be positioned symmetrically above the two lower rolls and shall be adjustable vertically to control the curvature of the workpiece. The two lower rolls shall be in a fixed position and both rolls shall be power driven.

3.4.4.3 Pinch-pyramid three roll. The pinch-pyramid, type III, forming machine shall have three roll s arranged in a symmetrical pyramid configuration. The top roll shall be symmetrically positioned above the lower rolls and shall be nonadjustable. The two lower rolls shall be independently adjustable angularly in relation to the top roll. All three rolls shall be power driven.

3.4.4.4 Double-pinch bending rolls. The double-pinch, type IV, forming machine shall have two pinch rolls and two bending rolls. The top pinch roll shall be mounted in a fixed position directly above the lower pinch roll. Bending rolls shall be mounted on each side of the lower pinch roll. The lower pinch roll and the two bending rolls shall be independently adjustable and tiltable hydraulically. The upper and lower pinch rolls shall be power driven.

3.4.4.5 Top roll. The top roll shall have means to release the pressure to allow removal of the finished product. Means shall be provided to return and secure the top roll in the operating position after removal of the finished product.

3.4.4.6 Pinch roll adjustments. The pinch roll shall be adjustable vertically or angularly based on the thickness of the workpiece. Unless otherwise specified, the adjustment shall be through either a separate motor or by hydraulic cylinders controlled from the operators station, When specified (see 6.2.1), the pinch roll on machine sizes 4-120 and smaller shall be manually adjustable by handwheel, handcrank, or similar means.

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TABLE I. Type I, initial-pinch sizes and capacities.

Size	Rolls				Material thickness	Main motor horsepower
	Top roll diameter (inches)	Effective working width, (inches)	Rolling speed, (fpm)			
			single 1/	variable		
2-24	2	24	16	-	20 ga.	1/3
2-36	2	36	16	-	22 ga.	1/3
3-24	3	24	15	-	12 ga.	3/4
3-36	3	36	16	-	14 ga.	3/4
3-42	3	42	16	-	15 ga.	3/4
3-48	3	48	16	-	16 ga.	3/4
3-60	3	60	16	-	20 ga.	3/4
3-80	3	80	16	-	16 ga.	1-1/2
4-36	4	36	15	-	10 ga.	1-1/2
4-42	4	42	16	-	11 ga.	1-1/2
4-48	4	48	15	-	12 ga.	1-1/2
4-60	4	60	15	-	14 ga.	1-1/2
4-72	4	72	15	-	16 ga.	1-1/2
4-84	4	84	15	-	16 ga.	1-1/2
4-96	4	96	15	-	20 ga.	1-1/2
4-120	4	120	15	-	20 ga.	3
6-36	6	36	15	-	1/4 in.	5
6-42	6	42	15	-	1/4 in.	5
6-48	6	48	16	-	1/4 in.	5
6-60	6	60	16	-	3/16 in.	5
6-72	6	72	16	-	10 ga.	5
6-84	6	84	16	-	10 ga.	5
6-96	6	96	16	-	12 ga.	5
6-120	6	120	16	-	14 ga.	5
6-144	6	144	16	-	1/16 in.	5

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TABLE I. Type I, initial-pinch sizes and capacities - Continued.

Size	Rolls				Material thicknes	Main motor horsepower
	Top roll diameter (inches)	Effective working width, (inches)	Rolling speed, (fpm)			
			single 1/	variable		
8-48	8	48	15	-	7/16 in.	7-1/2
8-60	8	60	15	-	3/8 in.	7-1/2
8-72	8	72	15	-	5/16 in.	7-1/2
8-96	8	96	15	-	7/32 in.	7-1/2
8-120	8	120	15	-	5/32 in.	7-1/2
8-144	8	144	15	-	9/64 in.	7-1/2
10-48	10	48	-	8 to 16	3/4 in.	10
10-72	10	72	-	8 to 16	9/16 in.	10
10-96	10	96	17	8 to 16	7/16 in.	10
10-120	10	120	-	8 to 16	3/8 in.	10
10-144	10	144	-	8 to 16	1/4 in.	10
10-168	10	168	-	8 to 16	3/16 in.	
12-48	12	48	-	6 to 12	7/8 in.	10
12-72	12	72	-	6 to 12	3/4 in.	10
12-96	12	96	17	6 to 12	5/8 in.	10
12-120	12	120	-	6 to 12	1/2 in.	15
12-144	12	144	-	6 to 12	3/8 in.	15
12-168	12	168	-	6 to 12	1/4 in.	15
12-192	12	192	-	6 to 12	3/16 in.	15
12-216	12	216	-	6 to 12	3/16 in.	15

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TABLE I. Type I, initial-pinch sizes and capacities - Continued.

Size	Rolls				Material thickness	Main motor horsepower
	Top roll diameter (inches)	Effective working width, (inches)	Rolling speed, (fpm)			
			single 1/	variable		
15-48	15	48	-	7 to 14	1-5/8 in.	25
15-72	15	72	-	7 to 14	1 in.	25
15-96	15	96	-	7 to 14	7/8 in.	25
15-120	15	120	-	7 to 14	3/4 in.	25
15-144	15	144	-	7 to 14	5/8 in.	25
15-168	15	168	-	7 to 14	1/2 in.	25
15-192	15	192	-	7 to 14	3/8 in.	25
15-216	15	216	-	7 to 14	13/32 in.	25
16-48	16	48	-	7 to 15	1-3/4 in.	30
16-72	16	72	-	7 to 15	1-1/8 in.	30
16-96	16	96	-	7 to 15	1 in.	30
16-120	16	120	-	7 to 15	7/8 in.	30
16-144	16	144	-	7 to 15	3/4 in.	30
16-168	16	168	-	7 to 15	5/8 in.	25
16-192	16	192	-	7 to 15	1/2 in.	25
16-216	16	216	-	7 to 15	3/8 in.	25
18-48	18	48	-	8 to 14	2 in.	40
18-72	18	72	-	8 to 14	1-3/8 in.	40
18-96	18	96	-	8 to 14	1-1/4 in.	40
18-120	18	120	-	8 to 14	1-1/8 in.	50
18-144	18	144	-	8 to 14	1 in.	50
18-168	18	168	-	8 to 14	13/16 in.	40
18-192	18	192	-	8 to 14	43/64 in.	40
18-216	18	216	-	8 to 14	1/2 in.	40
18-240	18	240	-	8 to 14	3/8 in.	40

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TABLE I. Type I, initial-pinch sizes and capacities - Continued.

Size	Rolls				Material thickness	Main motor horsepower
	Top roll diameter (inches)	Effective working width, (inches)	Rolling speed, (fpm)			
			single 1/	variable		
20-48	20	48	-	9 to 18	2-3/4 in.	60
20-72	20	72	-	9 to 18	1-3/4 in.	60
20-96	20	96	-	9 to 18	1-1/2 in.	60
20-120	20	120	-	9 to 18	1-1/4 in.	50
20-144	20	144	-	9 to 18	1-1/8 in.	50
20-168	20	168	-	9 to 18	15/16 in.	50
20-192	20	192	-	9 to 18	13/16 in.	50
20-216	20	216	-	9 to 18	43/64 in.	50
20-240	20	240	-	9 to 18	35/64 in.	50
22-72	22	72	-	10 to 18	2 in.	75
22-96	22	96	-	10 to 18	1-3/4 in.	75
22-120	22	120	-	10 to 18	1-1/2 in.	60
22-144	22	144	-	10 to 18	1-1/4 in.	60
22-168	22	168	-	10 to 18	1-1/16 in.	60
22-192	22	192	-	10 to 18	15/16 in.	60
22-216	22	216	-	10 to 18	7/8 in.	60
22-240	22	240	-	10 to 18	11/16 in.	60
24-72	24	72	-	11 to 19	2-1/4 in.	75
24-96	24	96	-	11 to 19	2 in.	75
24-120	24	120	-	11 to 19	1-3/4 in.	75
24-144	24	144	-	11 to 19	1-1/2 in.	75
24-168	24	168	-	11 to 19	1-1/4 in.	75
24-192	24	192	-	11 to 19	1-1/8 in.	75
24-216	24	216	-	11 to 19	1 in.	75
24-240	24	240	-	11 to 19	7/8 in.	75

1/ Single speed ±3 fpm.

Note: Requirements are not less than those shown. Where a range is shown, the requirements shall be not greater than the minimum to not less than the maximum figure shown in the table.

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TABLE II. Type II, pyramid sizes and capacities.

Size	Rolls				Material thickness	Main motor horsepower
	Top roll diameter (inches)	Effective working width, (inches)	Rolling speed, (fpm)			
			single 1/	variable		
3-40	3	40	20	-	10 ga.	1
3-50	3	50	20	-	10 ga.	1
4-40	4	40	20	-	3/16 in.	1-1/2
4-50	4	50	20	-	3/16 in.	1-1/2
4-60	4	60	20	-	10 ga.	1-1/2
4-80	4	80	20	-	14 ga.	1-1/2
5-60	5	60	20	-	1/4 in.	2
5-80	5	80	20	-	10 ga.	2
7-48	7	48	25	-	5/16 in.	7
7-60	7	60	25	-	1/4 in.	7
7-72	7	72	25	-	5 ga.	7
7-96	7	96	25	-	8 ga.	7
7-120	7	120	25	-	1/8 in.	7
10-48	10	48	-	8 to 16	3/4 in.	15
10-72	10	72	-	8 to 16	5/8 in.	15
10-96	10	96	-	8 to 16	1/2 in.	15
10-120	10	120	-	8 to 16	3/8 in.	15
10-144	10	144	-	8 to 16	1/4 in.	15
12-48	12	48	-	6 to 12	7/8 in.	20
12-72	12	72	-	6 to 12	3/4 in.	20
12-96	12	96	-	6 to 12	5/8 in.	20
12-120	12	120	-	6 to 12	1/2 in.	20
12-144	12	144	-	6 to 12	3/8 in.	20
13-48	13	48	-	7 to 14	1 in.	20
13-72	13	72	-	7 to 14	7/8 in.	20
13-96	13	96	-	7 to 14	3/4 in.	20
13-120	13	120	-	7 to 14	5/8 in.	20
13-144	13	144	-	7 to 14	1/2 in.	20

1/ single speed ± 3fpm

Note: Requirements are not less than those shown. Where a range is shown, the requirements shall be not greater than the minimum to not less than the maximum figure shown in the

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TABLE III. Type III, pinch-pyramid sizes and capacities.

Size	Rolls				Material thickness, (inches)	Main motor horsepower
	Top roll diameter (inches)	Effective working width, (inches)	Rolling speed, (fpm)			
			single 1/	variable		
8-72	8	72	22	0 to 23	3/8	7-1/2
8-96	8	96	-	0 to 23	5/16	7-1/2
8-120	8	120	-	0 to 23	1/4	7-1/2
8-144	8	144	-	0 to 23	1/4	7-1/2
10-72	10	72	22	0 to 23	5/8	10
10-96	10	96	-	0 to 23	1/2	10
10-120	10	120	-	0 to 23	3/8	10
10-144	10	144	-	0 to 23	5/16	10
12-72	12	72	-	0 to 23	7/8	20
12-96	12	96	-	0 to 23	3/4	20
12-120	12	120	-	0 to 23	5/8	20
12-144	12	144	-	0 to 23	1/2	20
14-72	14	72	20	0 to 23	1-1/8	25
14-96	14	96	20	0 to 23	1	25
14-120	14	120	22	0 to 23	3/4	25
14-144	14	144	22	0 to 23	5/8	25
14-168	14	168	22	0 to 23	1/2	25
16-72	16	72	20	0 to 23	1-1/4	40
16-96	16	96	-	0 to 23	1-1/8	40
16-120	16	120	20	0 to 23	1	40
16-144	16	144	20	0 to 23	3/4	40
16-168	16	168	22	0 to 23	5/8	40
18-72	18	72	15	0 to 23	1-1/2	40
18-96	18	96	15	0 to 23	1-3/8	40
18-120	18	120	15	0 to 23	1-1/4	50
18-144	18	144	15	0 to 23	1-1/8	50
18-168	18	168	15	0 to 23	1	50

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TABLE III. Type III, pinch-pyramid sizes and capacities - Continued.

Size	Rolls				Material thickness, (inches)	Main motor horsepower
	Top roll diameter (inches)	Effective working width, (inches)	Rolling speed, (fpm)			
			single 1/	variable		
20-72	20	72	15	0 to 23	2	75
20-96	20	96	15	0 to 23	1-3/4	75
20-120	20	120	15	0 to 23	1-3/8	75
20-144	20	144	-	0 to 23	1-1/4	75
20-168	20	168	15	0 to 23	1-1/8	75
22-144	22	144	15	0 to 23	1-1/2	80
22-168	22	168	15	0 to 23	1-1/4	80
24-144	24	144	12	0 to 23	2	110
24-168	24	168	12	0 to 23	1-1/2	110
26-168	26	168	12	0 to 23	1-3/4	110
27-168	27	168	12	0 to 23	2	125

1/ single speed ± 3fpm

Note: Requirements are not less than those shown. Where a range is shown, the requirements shall be not greater than the minimum to not less than the maximum figure shown in the table.

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TABLE IV. Type IV, double-pinch sizes and capacities.

Size	Rolls				Material thickness	Main motor horsepower
	Top roll diameter (inches)	Effective working width, (inches)	Rolling speed, (fpm)			
			single 1/	variable		
5-48	5	48	15	0-15	10 ga.	3
5-60	5	60	15	0-15	10 ga.	3
6-48	6	48	15	0-15	1/4 in.	3
6-72	6	72	15	0-15	10 ga.	3
8-48	8	48	15	0-15	3/8 in.	5
8-72	8	72	15	0-15	3/8 in.	5
8-96	8	96	15	0-15	1/4 in.	5
8-120	8	120	15	0-15	10 ga.	5
10-48	10	48	15	0-15	1/2 in.	7
10-96	10	96	15	0-15	3/8 in.	7
10-168	10	168	15	0-15	10 ga.	7
12-120	12	120	15	0-15	1/2 in.	20
12-240	12	240	15	0-15	10 ga.	20
14-60	14	60	15	0-15	1 in.	30
14-96	14	96	15	0-15	3/4 in.	30
14-144	14	144	15	0-15	1/2 in.	30
14-192	14	192	15	0-15	11/32 in.	30
16-60	16	60	15	0-15	1-3/8 in.	55
16-96	16	96	15	0-15	1 in.	55
16-144	16	144	15	0-15	11/16 in.	55
16-192	16	192	15	0-15	1/2 in.	55
16-240	16	240	15	0-15	5/16 in.	55

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TABLE IV. Type IV, double-pinch sizes and capacities - Continued.

Size	Rolls				Material thickness	Main motor horsepower
	Top roll diameter (inches)	Effective working width, (inches)	Rolling speed, (fpm)			
			single 1/	variable		
18-60	18	60	15	0-15	1-1/2 in.	70
18-96	18	96	15	0-15	1-3/16 in.	70
18-144	18	144	15	0-15	7/8 in.	70
18-240	18	240	15	0-15	3/8 in.	70
21-96	21	96	12	0-12	1-1/2 in.	120
21-144	21	144	12	0-12	1-3/16 in.	120
21-192	21	192	12	0-12	3/4 in.	120
21-240	21	240	12	0-12	1/2 in.	120
25-96	25	96	12	0-12	2-1/4 in.	200
25-192	25	192	12	0-12	1 in.	200
25-240	25	240	12	0.12	3/4 in.	200
27-120	27	120	12	0-12	2-3/8 in.	240
27-168	27	168	12	0-12	1-3/4 in.	240
27-240	27	240	12	0-12	7/8 in.	240

1/ single speed ± 3 fpm

Note: Requirements are not less than those shown. Where a range is shown, the requirements shall be not greater than the minimum to not less than the maximum figure shown in the table.

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3.4.4.7 Bending roll adjustments. Unless otherwise specified, the bending roll adjustment on types 1,11 and III machines having rolls up to 5 inches in diameter, shall be manually adjustable by handcrank or other similar means. The manual adjusting mechanism shall be located at each end of the roll. When specified (see 6.2.1), the bending roll adjustment on types, I, II, and III machines having rolls up to 5 inches in diameter shall be power operated. The bending roll adjustment on types I, II, and III machines with rolls over 5 inches in diameter and all type IV machines shall be power operated. With rolls in an idle or stop condition, the adjusting mechanism shall be capable of positioning the bending rolls to bend a full width and thickness sheet or plate material for the size machine specified in the applicable size and capacity table. Type I, III, and IV machines shall have prehending capabilities. The prebent edge flat shall be not greater than twice the material thickness. Adjustments for cone rolling shall be provided.

3.4.4.8 Roll position indicators. Roll position indicators shall be furnished to enable the operator to maintain parallel or tilt adjustment of the pinch roll. A mechanical indicator or dial, readable from the operating position, shall be furnished to enable the operator to accurately identify and set the position of the bending roll for duplicate work. When specified (see 6.2.1), the forming machines with power operated bending rolls shall have a digital readout for finite roll adjustments and duplication of workplaces. The digital readout shall indicate precise adjustment of lower rolls which are positioned remotely hydraulically or mechanically to provide parallelism and tilting of the lower rolls for cone rolling.

3.4.4.9 Roll bearings. Roll bearings shall be either bronze or antifriction type.

3.4.5 Tail hinge. The machine shall be equipped with either a manually, pneumatically, or hydraulically operated tail hinge. Unless otherwise specified (see 6.2.1), the tail hinge for machines having a top roll not greater than 5 inches in diameter shall be manually operated. Machines having a top roll over 5 inches in diameter shall be pneumatically or hydraulically operated. The tail hinge, when lowered, shall relieve the pinch allowing removal of the workpiece. When the tail hinge is returned to the operating position, the top roll shall return to the proper position to bend a duplicate workpiece. A locking device or other means shall secure the top roll and tail hinge in position.

3.4.6 Hydraulic system. The hydraulic system when furnished shall be complete, includlng all pumps, cylinders, motors, and pressure controls. Flow control valves shall be provided for control of both directions of rotation and roll speeds. Overpressure protection shall be provided in the high pressure line. A filter system shall be provided to assure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.4.7 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to NFPA 79. Each machine shall draw all of it's electrical power from a single 230/460 volt, 3 phase, 60 Hz circuit.

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Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in NFPA 79 for the disconnect fuse rating.

3.4.7.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2.1), each motor shall have a dripproof enclosure. Mechanical motor starter relays may be used with the main drive motor.

3.4.7.2 Single speed main drive. When single speed main drive motors are furnished, he electric motors shall have reversing and inching controls, and shall be provided with a magnetic brake.

3.4.7.3 Variable speed main drive. When variable speed main drive motors are furnished, the electric motors shall be variable speed, and shall be furnished with the necessary resistors and reversing controls.

3.4.7.4 Solid state components. Except for relays allowed in 3.4.7.1, solid state design shall be used throughout for electronic components.

3.4.7.5 Control circuit voltage. Auxiliary control circuits shall be isolated from he input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in tables I, II, III, and IV for the machine size specified. The bending capacities are based upon steel sheet and plate with a tensile strength of not less than 60,000 psi.

3.6 Performance. The machine shall be capable of bending sheet and plate into cylindrical shapes of the thickness and width shown in tables I, II, III, or IV for the size machine specified. The machine shall be capable of bending a finished cylinder of the full width and thickness rated capacity of the forming machine not greater than twice th top roll diameter.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, he requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, all equipment normally furnished with he manufacturer's standard commercial forming machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

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3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10. Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.10.1 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be securely attached to each forming machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means of mounting. The chart or plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.2 Nameplate. Unless otherwise specified (see 6.2.1), a nameplate shall be securely attached to each forming machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard forming machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number of Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Hour meter. Unless otherwise specified (see 6.2.1), the machine shall be equipped with an hour meter, The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of one hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

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3.12 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.13 Workmanship. Workmanship of the machine and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections, tests, and examinations shall be performed in the manufacturers designated indoor test area under the ambient temperature and the relative humidity and the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

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4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, and the inspection shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1. Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance test. Each machine shall bend a workpiece having a tensile strength of not less than 60,000 psi into a cylinder. The workpiece width and thickness shall be as shown by size in tables I, II, III, and IV. The inside diameter of the finished cylinder shall be not greater than twice the diameter of the top roll. Formed cylinders shall be round within five percent of the diameter. The prehending capabilities shall comply with 3.4.4.7.

4.7.3 Electromagnetic interference control test. Equipment requiring electromagnetic Interference control shall be tested for compliance with 3.3.9.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. Machines covered by this specification are intended for rolling steel sheet and plate into cylindrical shapes.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. First article, if required (see 3.1).

d. If the forming machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.1.1).

f. Dual calibrations (US and SI), if required (see 3.2.1.1.1).

g. Additional safety and health requirements, if required (see 3.2.5).

h. Centralized automatic lubrication system, if required (see 3.2.8).

i. Painting, if different (see 3.3.5).

j. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.9).

k. Single speed main drive, if required (see 3.4.3).

1. Lower roll circular grooves, if required (see 3.4.4).

m. Manual pinch roll adjustment, if required (see 3.4.4.6).

n. Power operated bending roll adjustment, if required (see 3.4.4.7).

o. Digital readout roll position indicators, if required (see 3,4.4.8).

p. Tail hinge operation, if different (see 3.4.5).

q. Electrical system and input power, if different (see 3.4.7).

r. Motor enclosure, if different (see 3.4.7.1).

s. Size and capacity, if different (see 3.5).

t. Additional performance features, if required (see 3.6.1).

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u. Standard equipment, if different (see 3.7).

v. Specify and fully describe optional equipment, if required (see 3.8).

w. Specify applicable repair parts, if required (see 3.9).

x. Lubrication chart or plate, if different (see 3.10.1).

y. Nameplate, if different (see 3.10.2).

z. Hour meter, if different (see 3.11).

aa. First article inspection, if different (see 4.4).

bb. Quality conformance inspection, if different (see 4.5).

cc. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required (see 5.1).

6.2.2 Contract data requirements Required technical data such as operators manuals, parts lists, wiring diagrams, foundation and anchor bolt plans and acceptance test reports should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those requirements that concern the forming machine itself. It does not require compliance with those OSHA requirements that concern the machine in its operating environment such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. OSHA limits the total hazard level of these hazards in the environment and does not limit the hazard level of individual machines in the environment. The procuring activity is advised to analyze the existing hazard requirements that will integrate the new machine into its future operating environment. The machine shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market. If specific point-of-operation guarding is required, the procuring activity should specify the exact configuration of the guard required, since the guard configuration is dependent on the size and configuration of the workplaces. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1 (g).

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6.5 Cross-reference of classification changes. Machine classification changes from previous issue of this specification are as follows:

a. Title changed from Forming Machine, Slip Roll, Initial Pinch Type Sheet and Plate Roll Dia. 2 Inches (50.8mm) to Forming Machines Sheet and Plate.

b, Type changes:

MIL-F-19426C	MIL-F-19426D
One type intial pinch	Type I – Initial-pinch, three roll Type II – Pyramid, three roll (New type) Type III – Pinch-pyramid, three roll (New type) Type IV – Double-pinch, four roll (New type)

c. The following sizes were deleted from MIL-F-19426C:

5-36	6-1/2-96	7-1/2-48	9-144	13-120	19-96
5-48	6-1/2-120	7-1/2-60	9-168	13-144	19-120
5-72	7-48	7-1/2-72	11-48	13-168	19-144
5-96	7-60	9-48	11-72	17-96	19-168
6-1/2-48	7-72	9-72	11-96	17-120	21-96
6-1/2-60	7-96	9-96	11-120	17-144	21-120
6-1/2-72	7-120	9-120	13-96	17-168	21-144

d. The following new sizes were added to the type I machine of MIL-F-19426D:

3-24	6-84	15-48	16-216	20-72	22-240
3-80	6-144	15-72	18-48	20-192	24-72
4-42	8-144	15-192	18-72	20-216	24-96
4-84	12-48	15-216	18-192	20-240	24-120
4-120	12-72	16-48	18-216	22-72	24-168
6-36	12-192	16-72	18-240	22-192	24-192
6-42	12-216	16-192	20-48	22-216	24-216
					24-240

e. All type II, III, and IV sizes are new sizes.

6.6 Subject term (key word) listing.

Cylinders, formed
Machine, bending, forming, shop

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Metal, fabrication
Plate, steel
Rolls, bending, forming
Shapes, cylindrical
Shop, fabricating
Steel, sheet, plate

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of changes.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Navy - YD
Air Force - 84
DLA - GS

User activities:
Army - SM, WV, MU
Navy - MC

Preparing activity:
DLA - IP

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-19426D	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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